Graph

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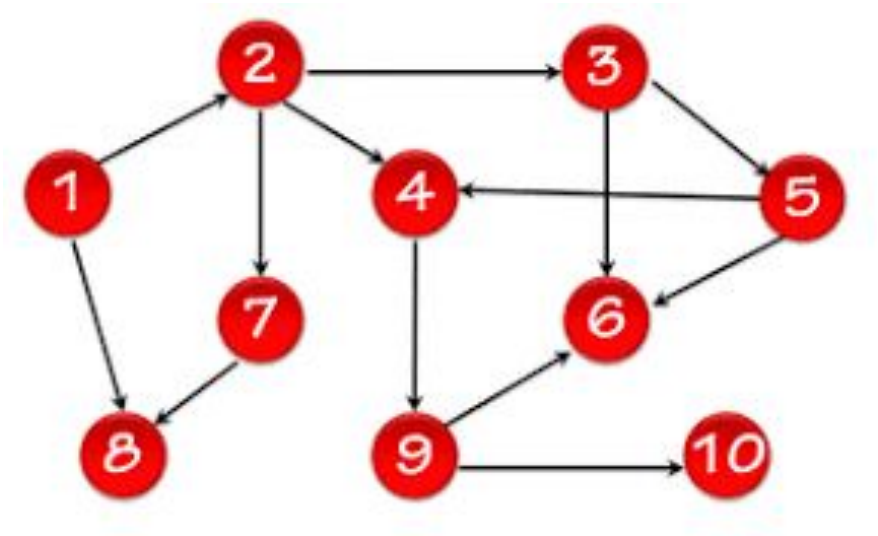
Theory

A **Graph** is a non-linear data structure consisting of a finite set of nodes (also called **vertices**) and a set of lines (called **edges**) that connect them. It is one of the most versatile structures in computer science because it can model almost any real-world relationship.

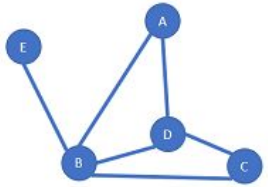
- **Vertices (V):** The individual points or nodes in the graph (e.g., cities on a map, users on a social network).
- **Edges (E):** The connections between the nodes (e.g., roads between cities, friendships between users).



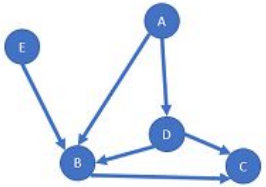
Key Classifications

A. Undirected vs. Directed

- **Undirected:** Edges have no direction. If Node A is connected to Node B, you can travel in both directions (like a two-way street).
- **Directed (Digraph):** Edges have a specific direction, usually represented by an arrow. If there is an edge from A to B, you cannot necessarily go from B to A (like a one-way street).



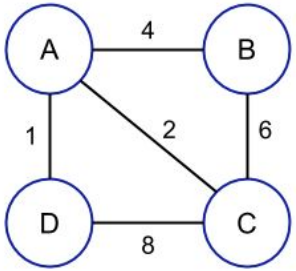
Undirected

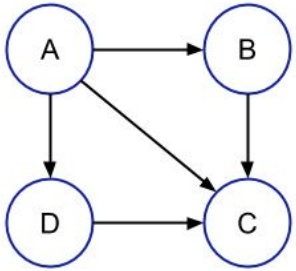


Directed

B. Weighted vs. Unweighted

- **Weighted:** Each edge has a numerical value (cost, distance, or time) associated with it.
- **Unweighted:** All edges are considered equal.

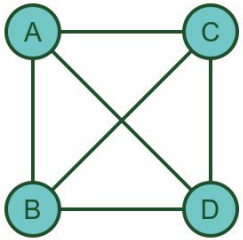




Representation in Code

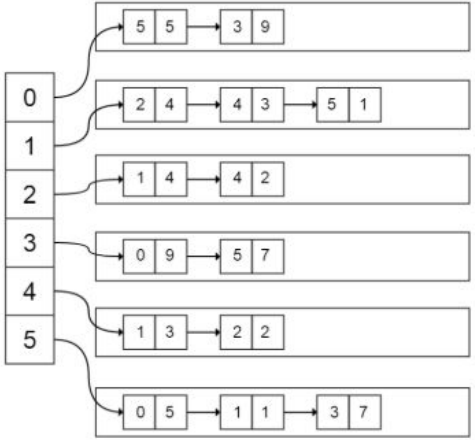
Adjacency Matrix: A 2D array where the cell at [i][j] indicates if there is an edge between node i and node j.

- *Pros:* Fast to check if an edge exists.
- *Cons:* Consumes more memory (V^2).



Complete graph

	A	B	C	D
A	0	1	1	1
B	1	0	1	1
C	1	1	0	1
D	1	1	1	0

Adjacency List: An array of lists(best maps). Each index represents a node, and its list contains all the nodes it is connected to.

- *Pros:* Memory efficient for "sparse" graphs (graphs with few edges).